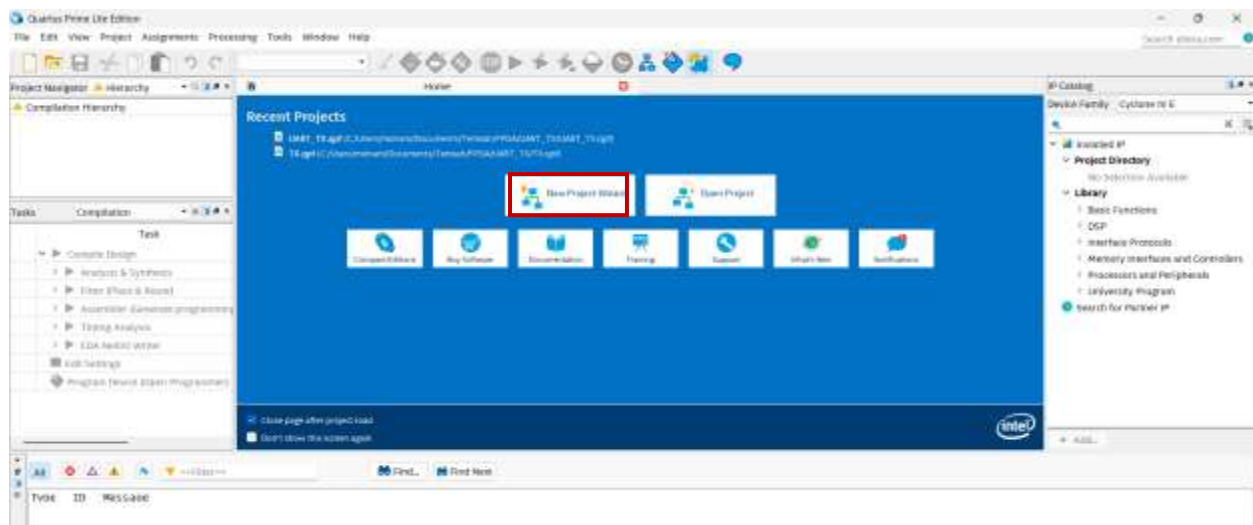


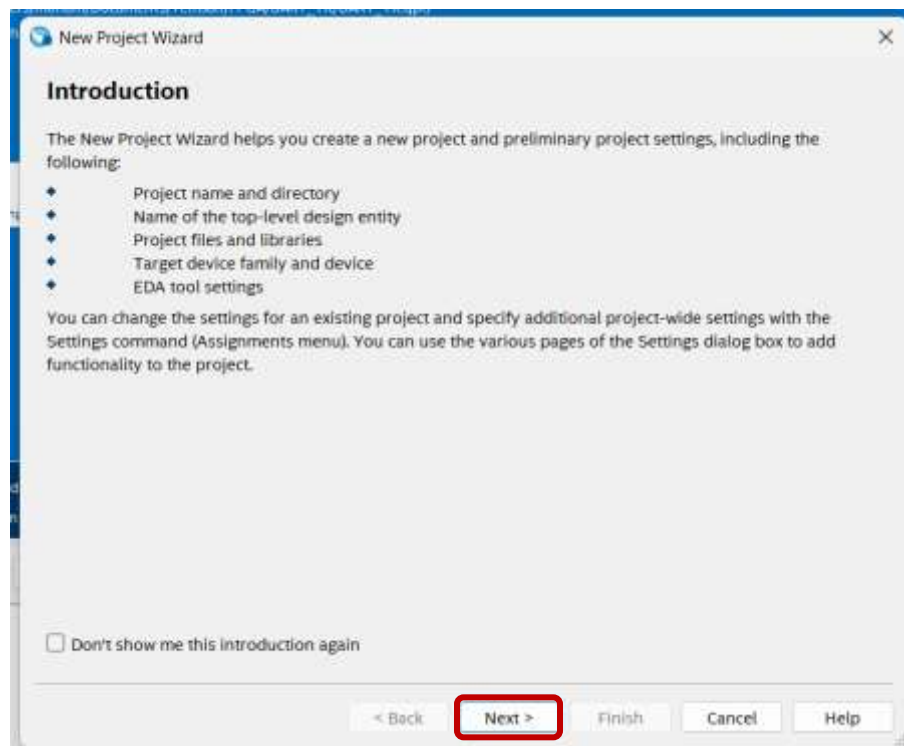
Lab1

Step 1: Project setup

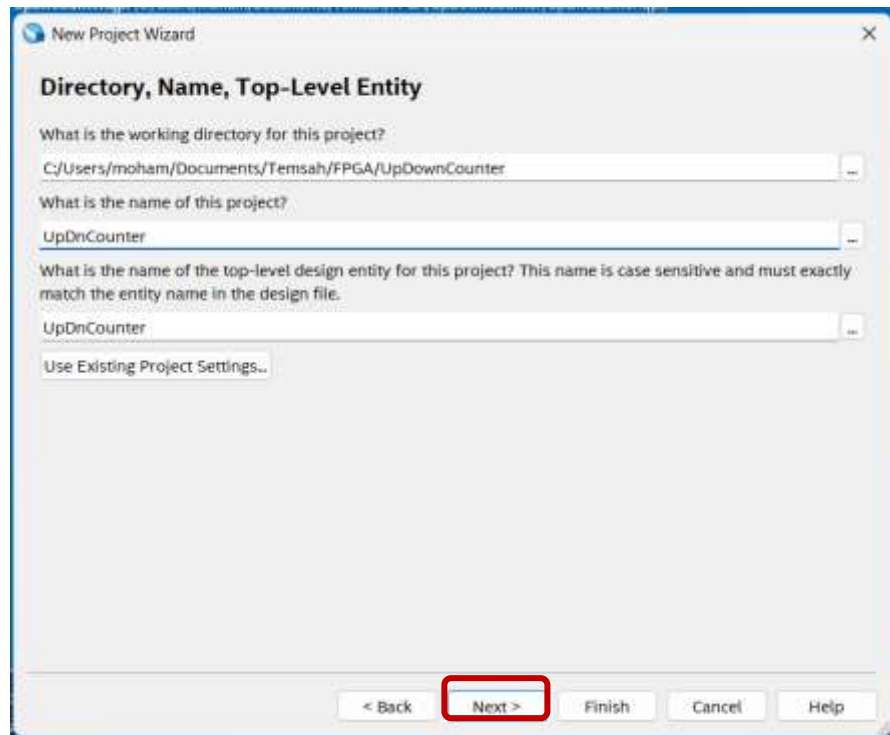
- Launch **Intel Quartus Prime Lite Edition** and click on **New Project Wizard** on the home landing page.
- Initializes a new design workspace, where the project directory, top-level entity name (UpDnCounter), target FPGA family (Cyclone IV), and EDA tools are defined.



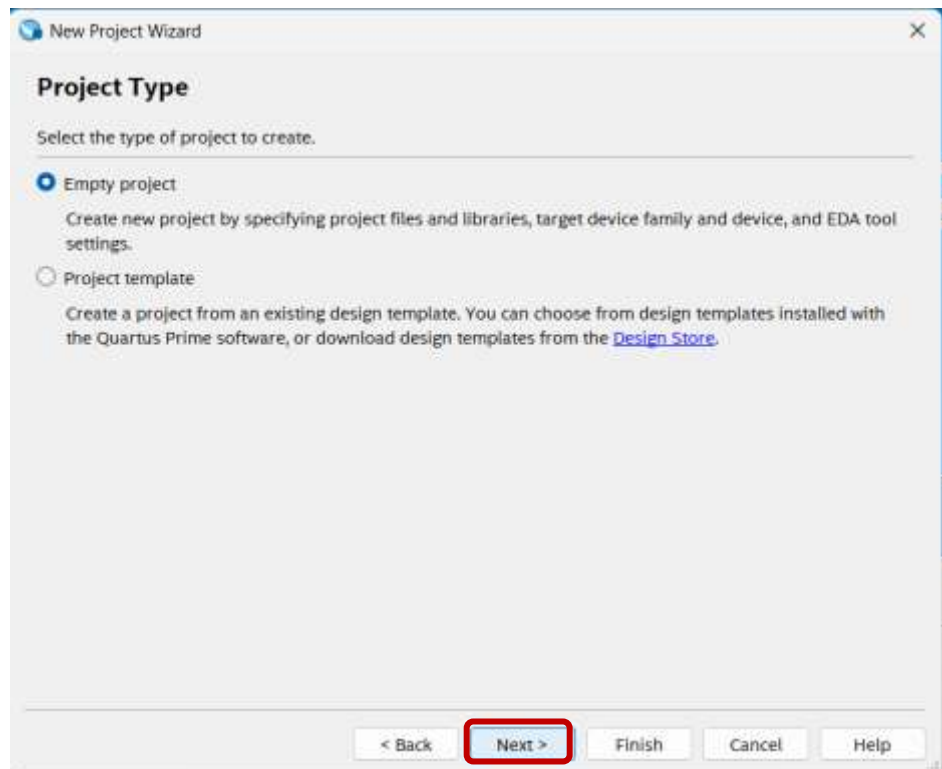
- Click **Next >** on the wizard introduction dialog to proceed with project configuration.



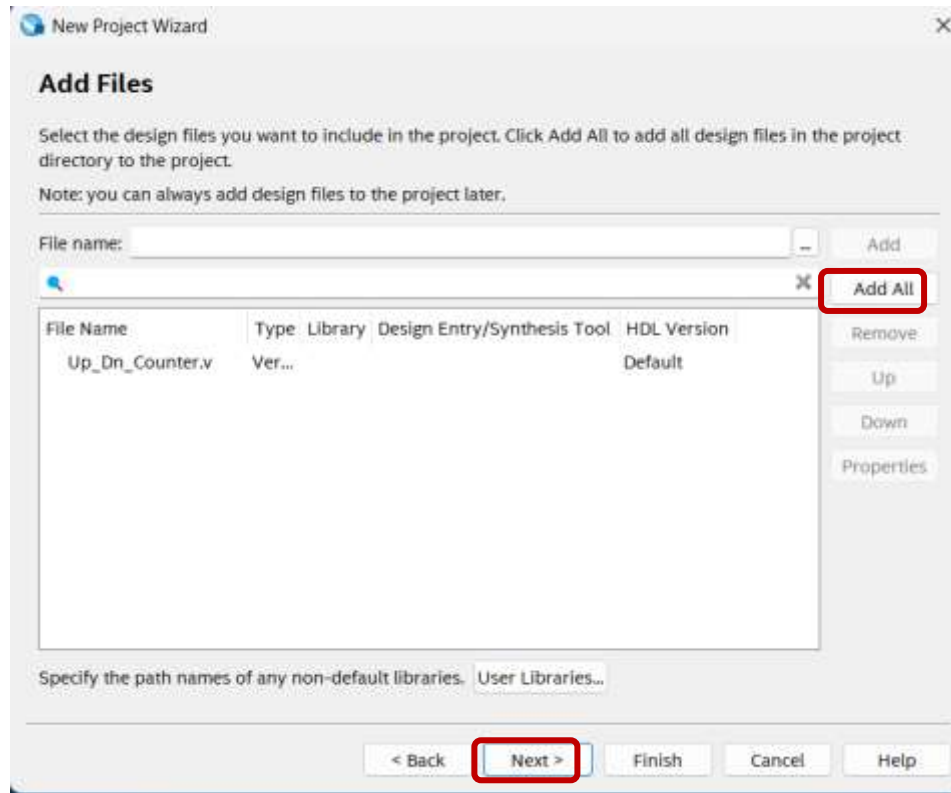
- Set the working directory path, name the project, and set the top-level entity name to UpDnCounter. Click **Next >**.



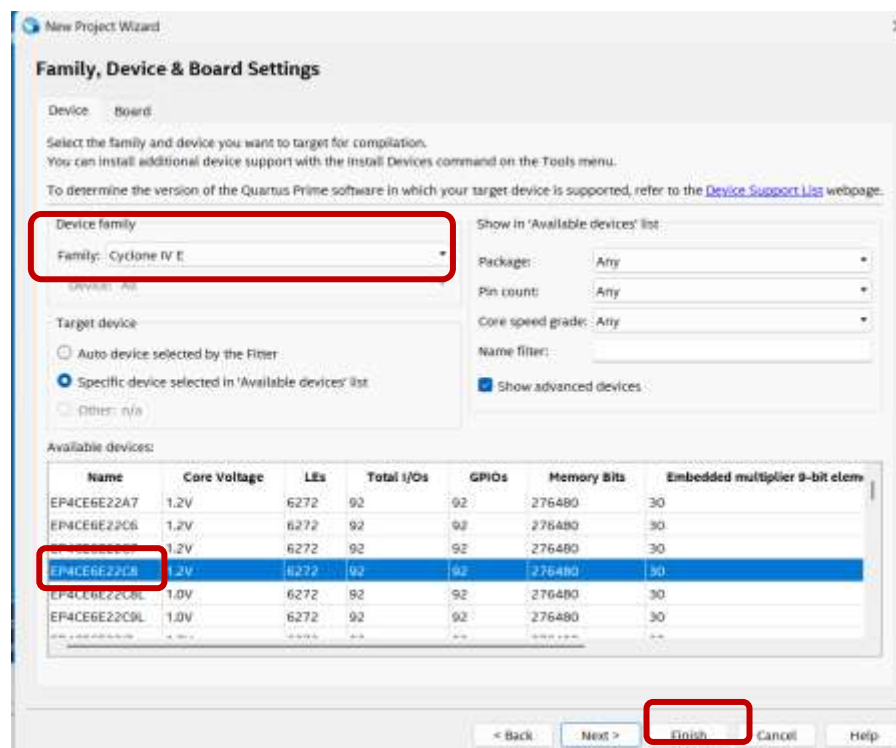
- Select **Empty project** and click **Next >** to create a fresh workspace for importing HDL source files and IP blocks.



- Add the Verilog design to the project by clicking **Add All**, then click **Next >**.

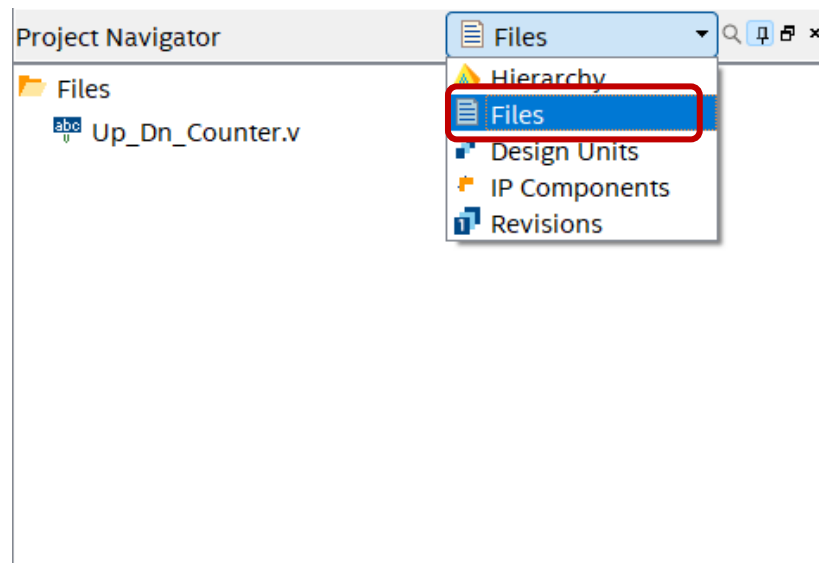


- Set the Device family to **Cyclone IV E**, select the target FPGA device **EP4CE6E22C8**, and click **Finish** to instantiate the workspace.

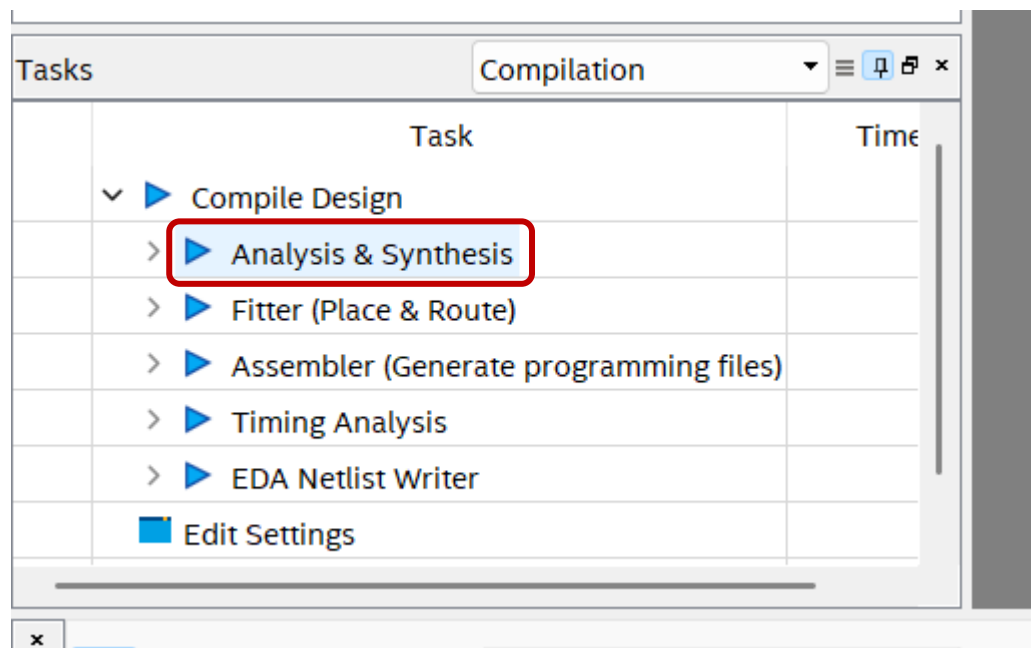


Step 2: Design Entry

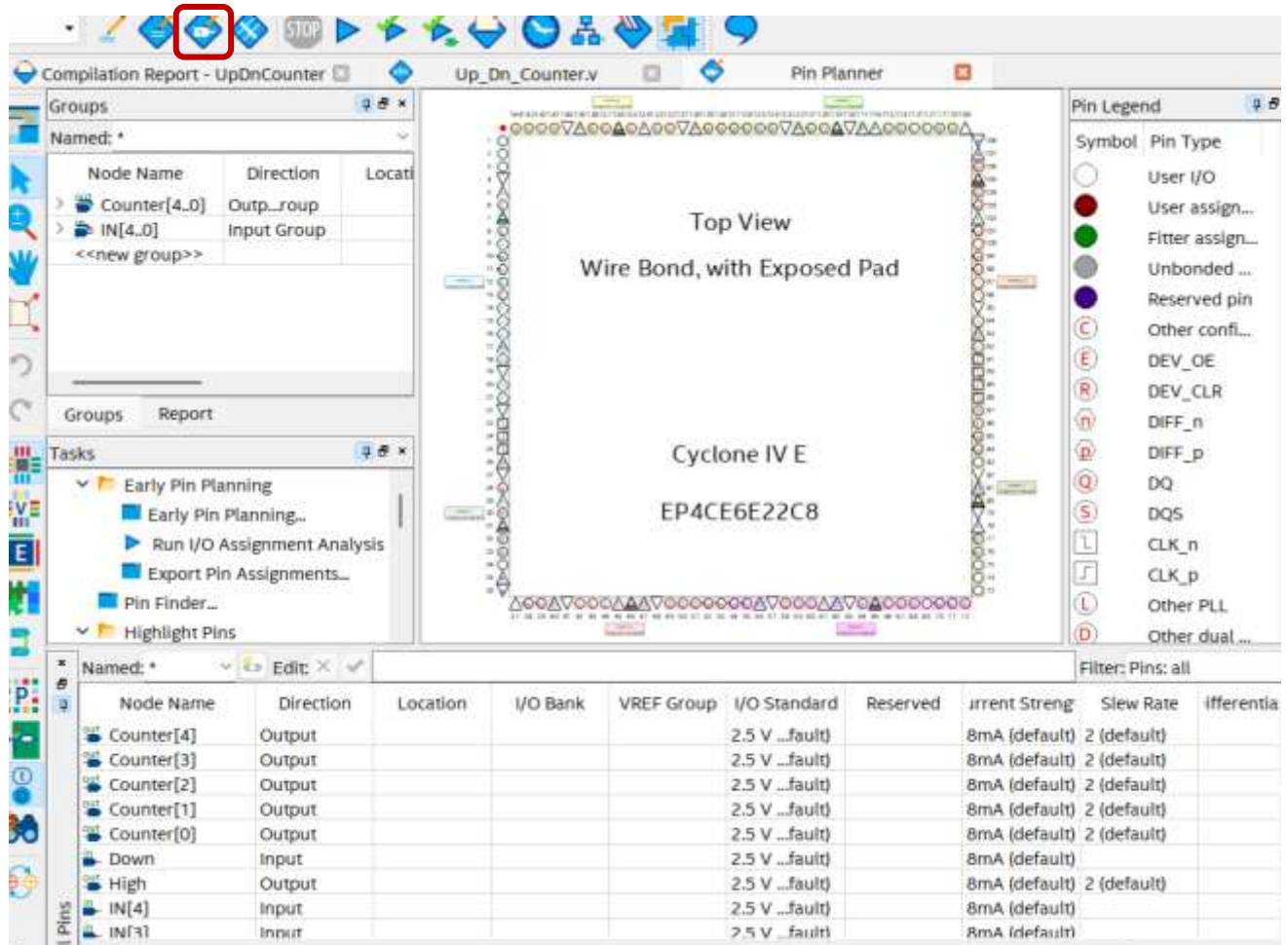
- Switch the Project Navigator dropdown view from **Hierarchy** to **Files** to display all added design source files.



- In the Tasks pane on the left, double-click **Analysis & Synthesis** to verify HDL syntax and synthesize the netlist into device logic elements.

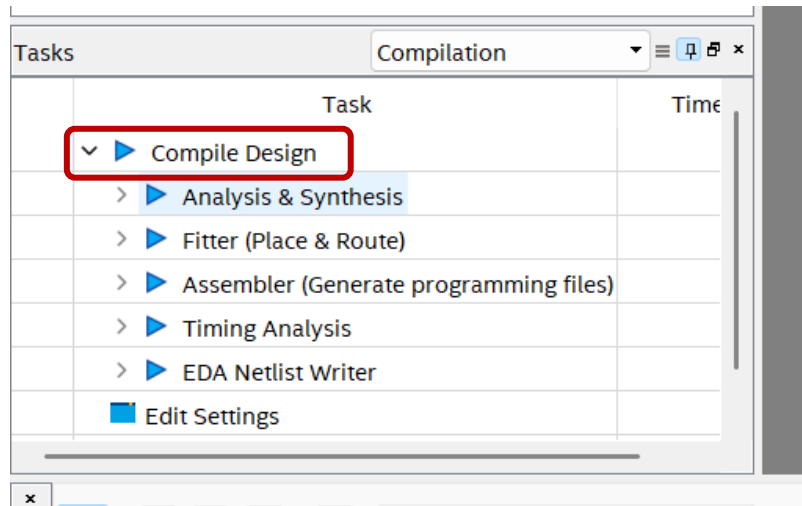


- Open **Pin Planner** by clicking the Pin Planner icon on the top toolbar

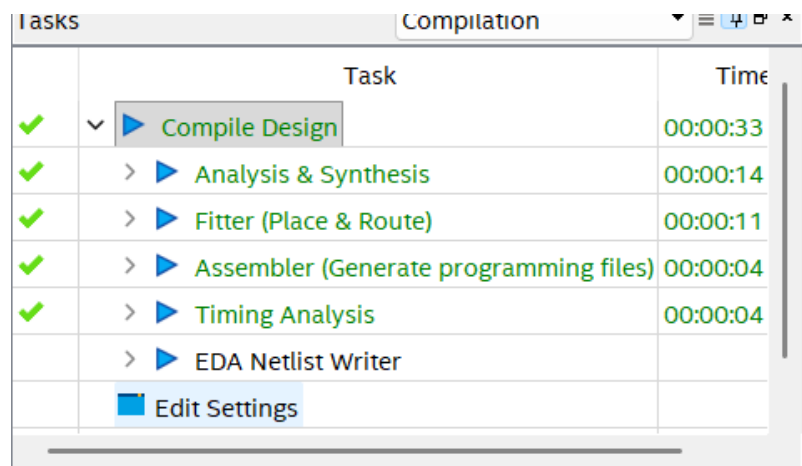
[illegible]

Step 4: Compilation Flow

- Double-click **Compile Design** in the Tasks pane to run the complete compilation flow, including Analysis & Synthesis, Fitter (Place & Route), Assembler, and Timing Analysis.

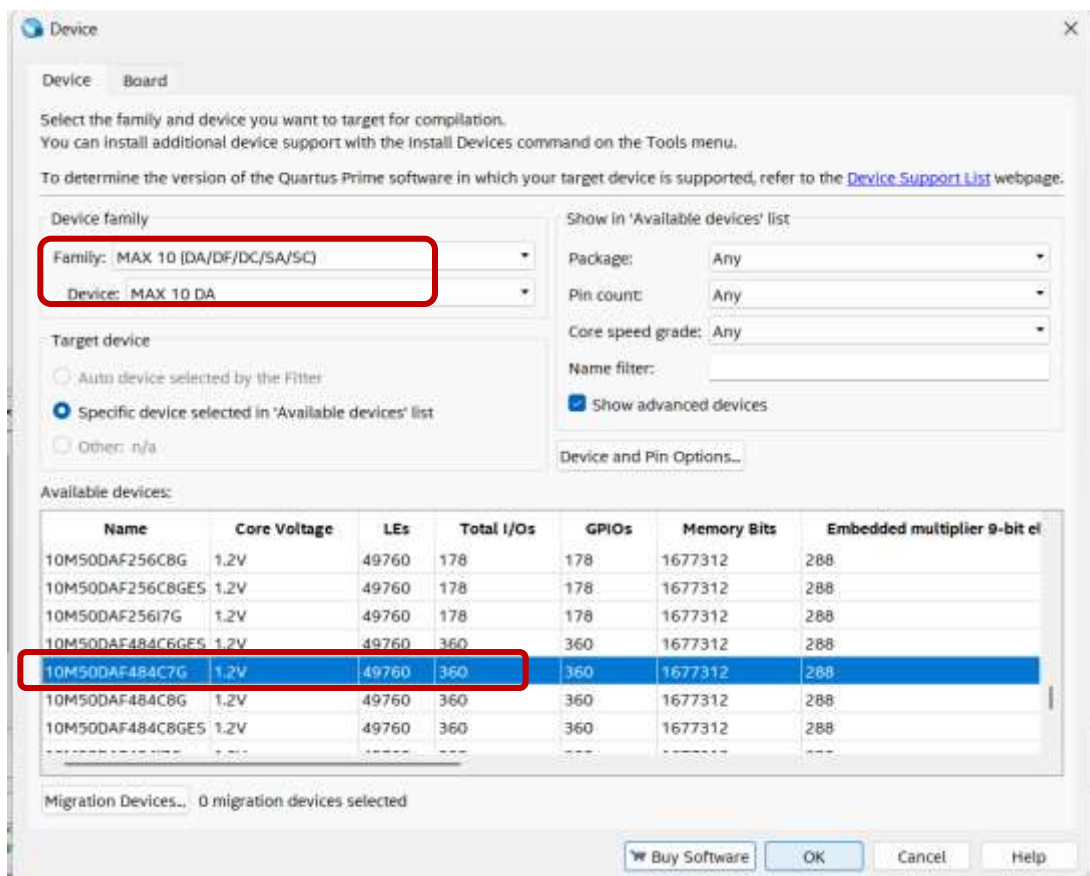
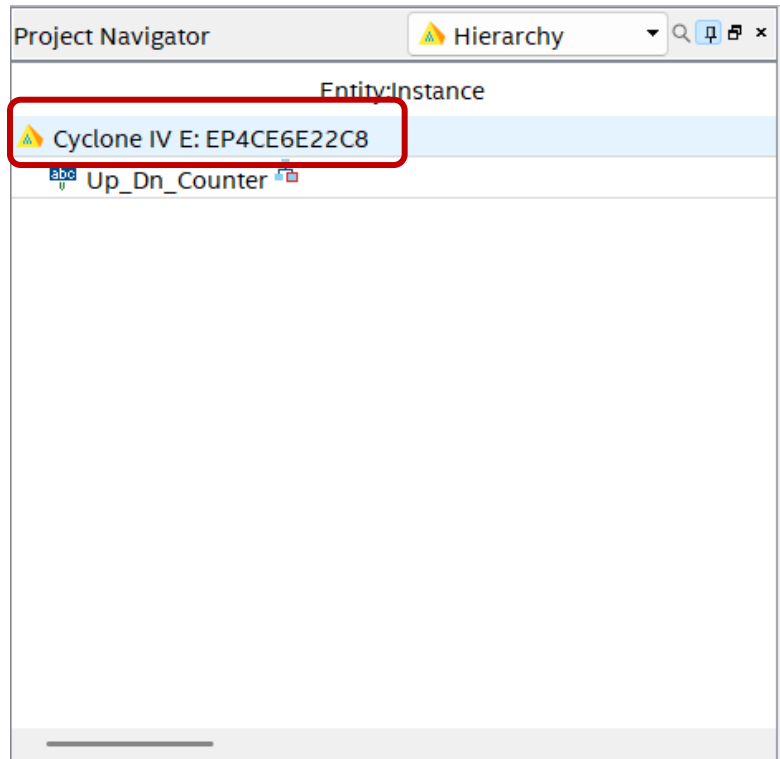


- Verify green checkmarks across all compilation stages

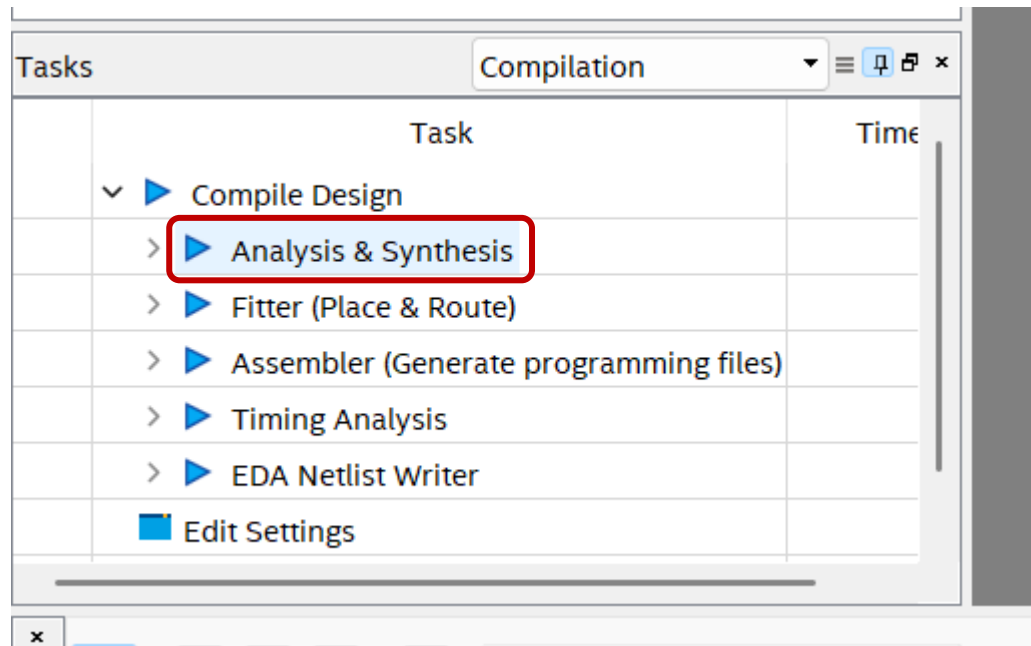


Step 5: Change Family

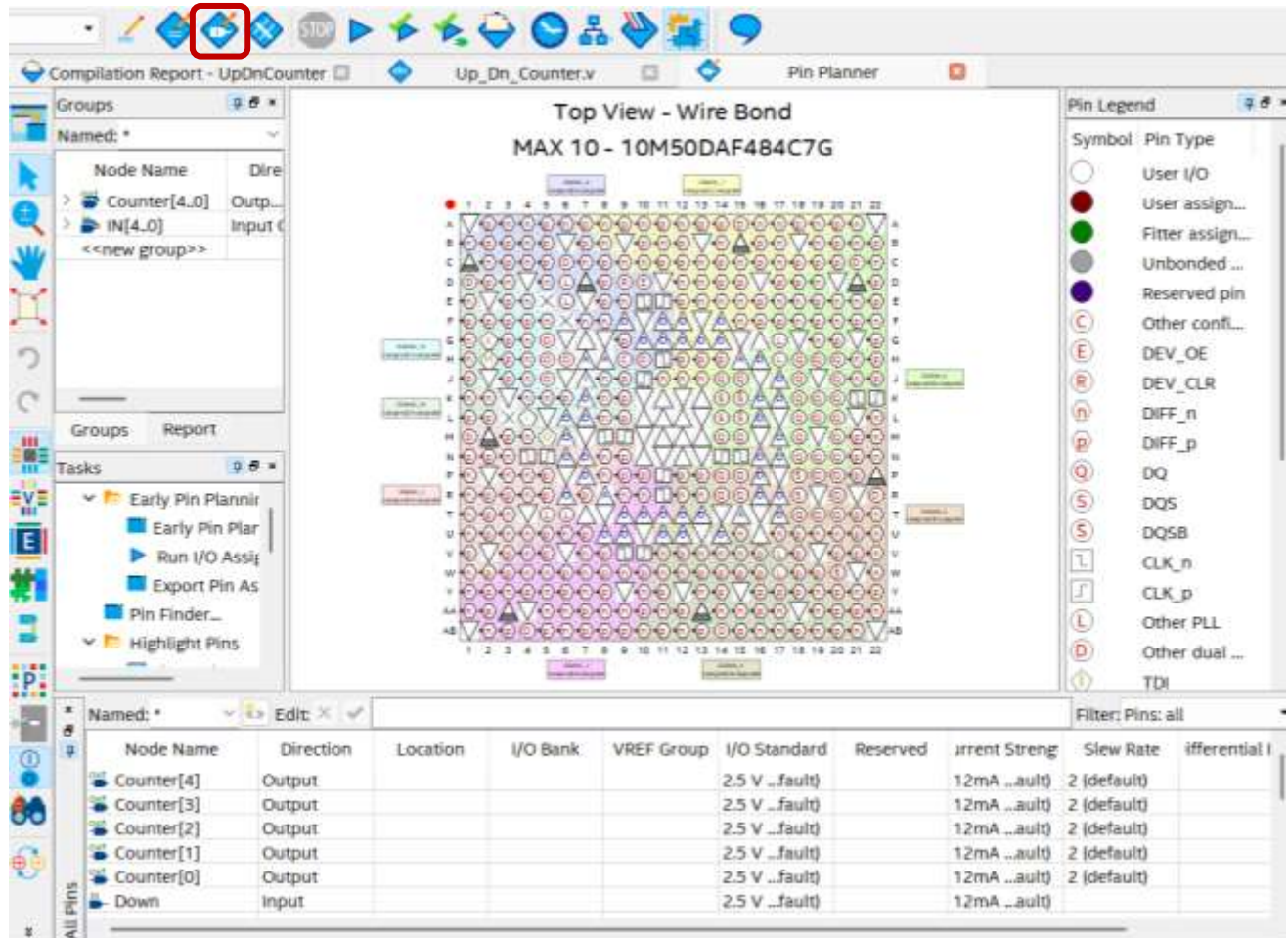
Now to change family return to hierarchy and double click family name



- In the Tasks pane on the left, double-click **Analysis & Synthesis** to verify HDL syntax and synthesize the netlist into device logic elements.



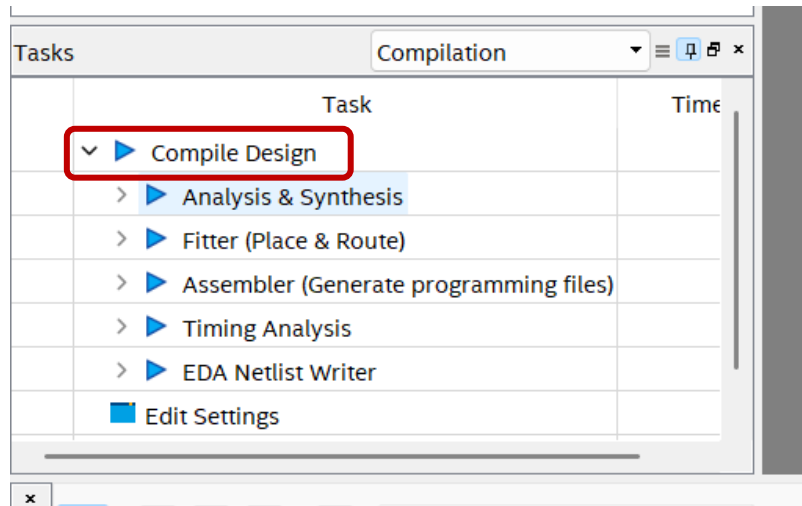
- Open **Pin Planner** by clicking the Pin Planner icon on the top toolbar



Named: *		Edit: X									
Node Name	Direction	Location	I/O Bank	VREF Group	I/O Standard	Reserved	Current Strength	Slew Rate	Differential Pair	Input Protection	
Counter[4]	Output	PIN_D13	7	B7_NO	2.5 V __fault		12mA __ault	2 (default)			
Counter[3]	Output	PIN_B10	7	B7_NO	2.5 V __fault		12mA __ault	2 (default)			
Counter[2]	Output	PIN_A10	7	B7_NO	2.5 V __fault		12mA __ault	2 (default)			
Counter[1]	Output	PIN_A9	7	B7_NO	2.5 V __fault		12mA __ault	2 (default)			
Counter[0]	Output	PIN_A8	7	B7_NO	2.5 V __fault		12mA __ault	2 (default)			
Down	Input	PIN_A14	7	B7_NO	2.5 V __fault		12mA __ault				
High	Output	PIN_B11	7	B7_NO	2.5 V __fault		12mA __ault	2 (default)			
IN[4]	Input	PIN_A12	7	B7_NO	2.5 V __fault		12mA __ault				
IN[3]	Input	PIN_C12	7	B7_NO	2.5 V __fault		12mA __ault				
IN[2]	Input	PIN_D12	7	B7_NO	2.5 V __fault		12mA __ault				
IN[1]	Input	PIN_C11	7	B7_NO	2.5 V __fault		12mA __ault				
IN[0]	Input	PIN_C10	7	B7_NO	2.5 V __fault		12mA __ault				
Load	Input	PIN_B12	7	B7_NO	2.5 V __fault		12mA __ault				
Low	Output	PIN_A11	7	B7_NO	2.5 V __fault		12mA __ault	2 (default)			
Up	Input	PIN_A13	7	B7_NO	2.5 V __fault		12mA __ault				
clk	Input	PIN_P11	3	B3_NO	2.5 V __fault		12mA __ault				
rst	Input	PIN_A15	7	B7_NO	2.5 V __fault		12mA __ault				
<<new node>>											

Step 6: Compilation Flow

- Double-click **Compile Design** in the Tasks pane to run the complete compilation flow, including Analysis & Synthesis, Fitter (Place & Route), Assembler, and Timing Analysis.



- Verify green checkmarks across all compilation stages

